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%% 1. Initial setup and synthetic piezoelectric sensor signal generation
fs = 10000;           % Sampling frequency (10 kHz)
t = 0:1/fs:3;         % Simulation time (3 seconds)

% Build a slip signal (high-frequency peaks at specific times)
% Assume slip occurs from 0.8 to 1.0 s and also from 2.2 to 2.5 s
slip_raw = zeros(size(t));
slip_idx1 = (t >= 0.8 & t <= 1.0);
slip_idx2 = (t >= 2.2 & t <= 2.5);

% Generate piezo oscillations (millivolt-level signals)
slip_raw(slip_idx1) = 0.015 * sin(2*pi*150*t(slip_idx1)) .* rand(1,
sum(slip_idx1));
slip_raw(slip_idx2) = 0.020 * sin(2*pi*200*t(slip_idx2)) .* rand(1, sum(slip_idx2));

% Add ambient noise (very low white noise)
noise = 0.002 * randn(size(t));
sensor_out = slip_raw + noise; % Sensor output (roughly -20 to +20 mV)

%% 2. Simulate the analog signal-conditioning circuit (gain and offset)
% Your circuit amplifies the signal and shifts it around 2.5 V (for single-supply
ADC)
Gain = 100;           % Amplifier gain
Offset = 2.5;         % DC offset voltage

analog_conditioned = (sensor_out * Gain) + Offset;
% Clamp voltage between 0 and 5 V (op-amp supply limitation)
analog_conditioned = max(min(analog_conditioned, 5), 0);

%% 3. Simulate the analog-to-digital converter (ADC)
bits = 8;             % 8-bit resolution
Vref = 5;             % ADC reference voltage
levels = 2^bits - 1;
resolution = Vref / levels; % ~19.6 mV per LSB

% Convert to digital (quantization)
digital_signal = round(analog_conditioned / resolution) * resolution;

%% 4. Controller (microcontroller) algorithm and calibration
% Since the signal is AC riding on 2.5 V, take absolute deviation from 2.5 V
signal_deviation = abs(digital_signal - Offset);

% Calibration (set the threshold)
Threshold = 0.8;       % If deviation exceeds 0.8 V, we consider it slip

% Slip detection
slip_detected_raw = signal_deviation > Threshold;

% Apply a debounce/hold so motors don't cut in/out abruptly
% Keeps the motor command high for a short time after detection

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motor_command = movmax(slip_detected_raw, 500); % Hold command for 50 ms

%% 5. Graphical user interface (GUI) and monitoring
figure('Name', 'Smart gripper monitoring UI', 'Position', [100, 100, 800, 600]);

% Plot 1: Raw sensor output
subplot(3,1,1);
plot(t, sensor_out * 1000, 'Color', [0.5 0.5 0.5]);
ylabel('Sensor voltage (mV)');
title('Raw piezoelectric sensor output (simulated)');
grid on;

% Plot 2: Conditioned/digitized signal and threshold calibration
subplot(3,1,2);
plot(t, digital_signal, 'b', 'LineWidth', 1.2); hold on;
yline(Offset, 'k--', 'Offset (2.5V)');
yline(Offset + Threshold, 'r--', 'Upper threshold');
yline(Offset - Threshold, 'r--', 'Lower threshold');
ylabel('ADC voltage (V)');
title('Digitized signal and threshold calibration');
ylim([0 5]);
grid on;

% Plot 3: Controller command to the gripper motor
subplot(3,1,3);
plot(t, motor_command, 'r', 'LineWidth', 2);
ylabel('Motor command');
xlabel('Time (s)');
title('Controller output (0 = normal, 1 = increase grip force)');
ylim([-0.2 1.2]);
yticks([0 1]);
yticklabels({'Normal', 'Grip Harder!'});
grid on;

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